University Echahid Hama Lakhdar, El-oued
Institute of Exact Sciences
Department of Computer Science
$2^{ed} Master$:
Artificial Intelligence and Data Science
Semester : 3.2025

Lab 8 : Object Detection using Faster R-CNN

Master Level – Computer Vision

Learning Objectives

After completing this lab, students will be able to :

- Understand improvements of Faster R-CNN over Fast R-CNN.
- Explore the role of Region Proposal Networks (RPN).
- Train and evaluate Faster R-CNN using PyTorch.
- Analyze anchors, proposals, RoI Align, and losses.
- Compare Faster R-CNN vs Fast R-CNN in terms of accuracy and speed.

Task 1 — Architecture Review

```
from torchvision.models.detection import fasterrcnn_resnet50_fpn
model = fasterrcnn_resnet50_fpn(pretrained=True)
print(model)
```

Questions

1. Identify :
   - Backbone (CNN + FPN)
   - RPN (Region Proposal Network)
   - RoI Align layer
   - Detection head (classifier + regressor)
2. How does RPN replace Selective Search ?
3. Where are anchors generated ?
4. Why does Faster R-CNN use RoI Align instead of RoI Pooling ?

Task 2 — Region Proposal Network (RPN)

```
from torchvision.models.detection.rpn import AnchorGenerator

anchor_gen = AnchorGenerator(
    sizes=((32, 64, 128, 256, 512),),
    aspect_ratios=((0.5, 1.0, 2.0),)
)
print(anchor_gen)
```

Questions

1. What are anchors and why use multiple sizes and ratios ?
2. Name the two RPN losses : classification and regression.
3. Define positive and negative anchors :

$$IoU \geq 0.7, \qquad IoU \leq 0.3$$

4. Visualize anchors and describe what you observe.

Task 3 — Extracting Region Proposals

```
import torchvision
from PIL import Image
from torchvision import transforms
import torch

img = Image.open("samples/dog.jpg")
x = transforms.ToTensor()(img).unsqueeze(0)

model = torchvision.models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
model.eval()

with torch.no_grad():
    det = model(x)

print(det[0]["boxes"][:5])
```

Questions

1. How many proposals does RPN generate before NMS ?
2. How many remain after NMS ?
3. Compare RPN proposal scores vs final detection scores.
4. Compare RPN speed vs Selective Search from Lab 7.

Task 4 — RoI Align Layer

```
import torch
from torchvision.ops import roi_align

feature_map = torch.randn(1, 256, 60, 60)
rois = torch.tensor([
    [0, 10, 10, 40, 50],
    [0, 20, 15, 50, 60]
], dtype=torch.float)

pooled = roi_align(feature_map, rois,
                   output_size=(7,7),
                   spatial_scale=1/16)

print(pooled.shape)
```

Questions

1. Why is RoI Align more accurate than RoI Pooling ?
2. What is the purpose of the $7 \times 7 \times 256$ output ?
3. Increase output size to $(14, 14)$ and observe memory/computation changes.

Task 5 — Multi-Task Loss Function

The Faster R-CNN loss is :

$$L = L_{\text{RPN cls}} + L_{\text{RPN loc}} + L_{\text{ROI cls}} + L_{\text{ROI loc}}.$$

```
import torch
import torch.nn.functional as F
```

```
cls_target = torch.tensor([1, 0, 2])
bbox_target = torch.randn(3, 4)
cls_pred = torch.randn(3, 3)
bbox_pred = torch.randn(3, 4)

L_cls = F.cross_entropy(cls_pred, cls_target)
L_loc = F.smooth_l1_loss(bbox_pred, bbox_target)

L_total = L_cls + L_loc
print("Loss:", L_total.item())
```

Questions

1. Why combine losses from RPN and ROI head ?
2. Why does Smooth L1 loss work better than MSE ?
3. What happens if RPN classification loss becomes unstable ?

Task 6 — Training Faster R-CNN

```
import torchvision

dataset = torchvision.datasets.PennFudanPed(
    root="data",
    transforms=transform_train
)
```

Questions

1. What learning rate gives stable training ?
2. Which layers are fine-tuned ?
3. Plot :
   - RPN classification loss
   - RPN regression loss
   - ROI classification loss
   - ROI regression loss
4. Evaluate mAP on test images.

Task 7 — Visualization and Analysis

```
with torch.no_grad():
    det = model([transforms.ToTensor()(img)])[0]
```

Questions

1. Apply NMS with thresholds 0.3, 0.5, 0.7. Compare results.
2. Compare failure cases between Fast R-CNN and Faster R-CNN.
3. Identify classification and localization errors.
4. How does RPN reduce the errors seen with Selective Search ?

Final Comparative Questions (Lab 7 → Lab 8)

1. Replace RoI Pooling with RoI Align. How does mAP change ?

2. Reduce proposals :

$$2000 \rightarrow 300 \rightarrow 100$$

   Evaluate speed and accuracy.
3. Compare freezing vs fine-tuning backbone layers.
4. Apply Faster R-CNN to the vignette dataset and report performance.